

Electronic Maintenance Support System (EMSS) Kit, Attachment 1

Purpose

To provide the technical specifications and additional considerations required for a ruggedized laptop/tablet solution to be purchased and utilized as a part of an EMSS Kit.

Technical Specifications

The following tables provide the minimum acceptable technical specifications for a ruggedized laptop/tablet with kit solution. The laptops/tablets are also known as an Electronic Maintenance Device (EMD) and will be referred to as such throughout this document.

Ruggedized EMD and Kit:

Attribute	Minimum Requirement
* Central Processing Unit (CPU)	3.3 Gigahertz (GHz) or faster with 8 efficient cores or more cores capable of 16 or more threads on a compatible 64-bit processor
* Random Access Memory (RAM)	32 GB DDR4 1866 MT/s; Expandable to minimum of 64 GB

* Graphics Processing Unit (GPU)	Compatible with DirectX 12 or later with WDDM 2.0 Video Output: HDMI Base Clock: 1515 Megahertz (MHz) Boost Clock: 1710 MHz Memory Clock: 1750 MHz, 14 Gigabits per second (Gbps) effective Pixel Rate: 109.4 GPixel/s Texture Rate: 314.6 GTexel/s FP16 (half) performance: 20.14 TFLOPS (2:1) FP32 (float) performance: 10.07 TFLOPS FP64 (double) performance: 314.6 GFLOPS (1:32) Memory Size: 4 GB Memory Type: GDDR6
* Storage	Capacity. The EMD shall include a Solid State Drive (SSD) Hard Drive that is no less than 1 Terabyte (TB) Removable. The EMD hard drive shall be removable from the EMD using common tools available in standard issue Marine Corps sets, chests, and kits Hard Drive Interface. The EMD Hard Drive interface (e.g., Serial Advanced Technology Attachment (SATA) III) shall support data transfer rates of at least 4.8 GB per second. The hard drive solution is NOT required to meet/use OPAL specifications as full disk encryption will be a software based solution provided by the Government. The vendor is NOT responsible for providing encryption software.
Trusted Platform Module	Contain a Trusted Platform Module (TPM) hardware chip, version 2.0 or higher, with associated Computer Root of Trust Measurement.
Display Adapter	Display Adapter - Full High Definition (1080P) display that is greater than 10.1" diagonally, 8 bits per color channel XGA daylight/outdoor readable screen; Touch screen; Shall support 1920x1080 resolution or better.

Communications Ports	Network port - Gigabit Ethernet network interface USB Ports - A minimum of four USB ports, two of which shall be a USB A type port 2.0 or above, and at least one USB C type port 3.0 or above. The EMD shall have an internal High-Definition Multimedia Interface (HDMI) port. Serial Port True RS-232
Smart Card Reader	Yes
Wireless	Wireless adapter shall be at least 802.11ax compatible. (Optional) Able to connect to Light Fidelity (Li-Fi) 802.11bb through on board adapter or with an external adapter.
Audio	The EMD shall have internal line-in, stereo headphone, and microphone connectors. A combined port for all three may be used. Headphone Connector. The EMD headphone connector shall accept a three-section 3.5 millimeter (mm) stereo phone plug consisting of a tip, ferrule, and barrel. Auto-Switch Output. The EMD headphone connector shall be of the "break-to-make" type that automatically breaks the circuit to internal speaker(s) and connects the stereo amplifier outputs to the external speakers (in headphone) when the headphone plug is inserted. Internal Speaker Audio Switch. The EMD shall provide headphone connectors that switch the stereo output back to the internal speaker(s) when the external headphone plug is removed. Sound Projection. The EMD internal speakers shall have sound pressure output of at least 80 decibel (dB) from 250 Hertz (Hz) to 5 Kilohertz (kHz) measured at 2 feet from the front of the display. The EMD shall be capable of accessing headphone/headset and speaker audio via USB connection.
Input Devices	82-key water resistant keyboard; Mouse Touchpad; Stylus Pen; Standard on-board camera with built-in lenses cover. The EMD shall contain input devices (i.e., keyboard and mouse, touch screen and stylus, voice and/or any other new input devices) that allow interaction with the system and can be changed while the device is still powered on (hot swapped) as per changing objectives.

Power	AC Adapter 100-240V 50/60 Hz; The AC adapter shall be a part of the EMD CLIN and its cost unless ordered in a separate CLIN as a spare.
* Battery	Shall be charged at 100 V – 240 V. The primary battery on the EMD must provide a runtime of 8 hours or more. At least one battery shall be a part of the EMD CLIN and its cost. This battery will be referred to as the primary battery.
NATO Adapter	Shall be used at an input of 12 V – 32 V and output must be compatible with the input rate of the EMD
Weight	The EMD shall have a maximum weight of 9 lbs. (37 lbs. with backpack and ancillary equipment) and meet DoD Human Engineering Design Criteria Standards (MIL-STD-1472H) Objective weight for the EMD is 5 lbs. (25 lbs. with backpack and ancillary equipment) or lighter.
Environmental Specifications	Comply with equipment ruggedization in accordance with DoD Environmental Engineering Considerations (MIL-STD-810G) tests for operational temperature, non-operational temperature, rain, humidity (operational), salt fog, dust/sand, vibration, and shock (drop and functional).
Ruggedized	See Environmental Specifications in the above row of this table.
Software	Platform must be compatible with 32 and 64 bit applications; Be Windows 11 Secure Boot compliant and support Unified Extensible Firmware Interface (UEFI) boot mode.
Mean Time Between Failure (MTBF)	Minimum to 500 hours before maintenance action is required.
Power On Self-Test Capability	The EMD shall have a self-test capability to verify its operational status prior to use. Verification time should be no more than 1 minute under normal operations.
* Warranty	Minimum 3 years warranty support for parts and labor; 24x7 Monday-Sunday phone support; Mail-In Service
Customization Notes	N/A

Additional Considerations and Requirements

Accessories

Each of the following accessory items or equivalent shall also be provided to support use of the proposed ruggedized laptop/tablet solution. All other accessories and the EMD shall be able to be stored within the Back Pack being requested/proposed as to provide all items as a finalized kit to the USMC Fleet.

- **Universal Back Pack, Computer –**
 - The backpack shall be large enough to hold the EMD and all listed accessories.
 - The back pack shall have at minimum 3 separate compartments to allow for organizing and separating the different accessories.
 - The back pack shall be constructed primarily of Cordura, Kordra, Ballistic Nylon, Dyneema or a combination of those materials.
 - The back pack shall have a minimum of 6 stitches per inch on seams.
 - The back pack shall utilize YKK zippers.
 - The back pack shall have webbing stitched on the outside of the pack in a MOLLE configuration.
- **Adapters, Power –** There shall be 2 power adapters in the EMD kit.
 - A standard A/C power adapter that shall be included as a part of the EMD CLIN
 - AC Adapter 100-240V 50/60 Hz;
 - A NATO power adapter that will be its own separate CLIN
 - Shall be able to be used at an input of 12 V – 32 V
 - Shall have an output range that will support the laptop/tablet being proposed.
- **Battery Charger –** There shall be a standalone battery charger that will allow for charging the laptop/tablet batteries externally from the system or for charging spare batteries.
 - Size is a consideration for this item. It must be able to fit in the back pack with all other accessories and the EMD.
 - Threshold for the battery charger is to charge one to four batteries that are compatible with the EMD.
 - Objective for this charger is to charge the number of batteries that exist onboard the EMD.
 - i.e. If the EMD had 1 battery in the keyboard and 1 battery on the tablet, the charger shall be capable of charging 2 batteries at a time.
- **Ethernet Cable –** 20 foot Ethernet cable CAT 6 with RJ45 connectors on both ends. **Shall not** be red in color.
- **Strap Assembly (Optional) –** There shall be a strap assembly or similar solution that allows for the user to securely hold the EMD with one hand and allow for typing, stylus use, or tablet use.
 - For an example of a previous fielded solution, see P/N: TBC20HDSTP-P
- **External USB DVD-ROM/CD-RW Drive:** USB DVD/CD-RW 8X/24X write speeds or better drive with captive power and data cables.
- **Expansion Laptop Battery / additional rechargeable expansion battery –** The EMD shall come with 1 battery that has a runtime of 8 hours as a part of the EMD CLIN. The expansion battery shall also be

compatible with and be able to be charged by the external battery charger. The expansion battery coupled with the primary battery shall provide additional runtime beyond the 8 hours specified for the primary battery. The expansion battery shall be charged at 100 V - 240 V. by the same charger used to charge the primary battery

- When considering backpack space, this battery shall be installed in the EMD system and shall not take up any additional space.

Spares

Hard Drives:

The proposed solution shall provide the option for a quantity up to 200 spare hard drives that match the initial hard drive in the EMD. This will be an internal hard drive.

- Up to 200 (qty) Solid State Drive (SSD) Hard Drive that is no less than 1 Terabyte (TB)

Batteries:

The proposed solution shall provide the option for a quantity of up to 50 primary batteries used in the EMD proposed solution.

A/C Power Adapters:

The proposed solutions shall provide the option of up to 200 spare A/C power adapters that match the original power adapter part provided within the EMD CLIN.

Item Unique Identification (IUID) Branding

Prior to delivery to the Government and based on DoD requirements each ruggedized laptop/tablet shall receive an IUID tag, which as indicated by the name uniquely identifies each device. The basic construct of an IUID tag is a collection of several device identifiers and a 2D barcode. Sample information for an IUID tag is provided below for reference and the solution provider (vendor) must contact Automatic Test System Team Lead (ATS TL) to verify the information and values to be included prior to tagging each ruggedized laptop prior to shipment. The IUID Barcode or QR code shall be in construct 1 format.

NSN: *(xxx-xx-xxx-xxxx)*

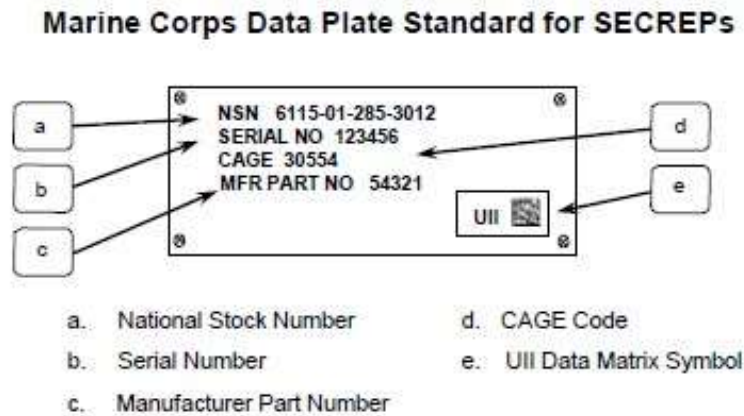
SERIAL NUMBER: *(Serial Number of each Device)*

MFR CAGE: 12255 *(USN NSWC Crane, IN)*

Manufacturer Part Number *(xxxxx)*

Example is below.

Item Identification and Valuation in accordance with DFARS 252.211-7003 is required for this line item.



NOTE: The above format is mandatory for Marine Corps secondary reparable and contains the maximum number of data attributes.

Figure 1 - Sample IUID Tag

BRANDING:

Program Branding (Logo) Prior to delivery to the Government a single EMSS Program logo shall be securely affixed to each ruggedized laptop/tablet. The logo shall be oriented so that when the EMD is facing a user and the lid is closed the logo appears centered and readable. The logo shall be attached so that it is clearly visible and will not delaminate or come unattached when contacted with reasonable amounts of water, dirt, debris, grease, liquid or vibration.